Stated in calculus terminology, velocity, $v = \frac{ds}{dt}$, is the rate of change of position with time, and acceleration, $a = \frac{dv}{dt}$, is rate of change of velocity with time.

The acceleration to be negative because if you take the direction of movement to be positive, then for a bike that is slowing down, its acceleration vector will be oriented in the opposite direction of its motion (retardation).

Given that the retardation of the car is 8 meter/second^2.

$$\text{Therefore, } a = \frac{dv}{dt} = -8 \text{ meter/second}^2.$$

$$\text{Therefore, } v = \int a\,dt = \int -8dt = -8t + c_1$$

$$v = -8t + c_1.$$

When the brakes are applied,

$$t = 0, \text{ and } v = 24\text{m/s}.$$

$$\text{So, } 24 = -8(0) + c_1 \Rightarrow c_1 = 24$$

$$\text{Therefore, } v = -8t + 24.$$

$$\text{That is, } \frac{ds}{dt} = -8t + 24.$$

It is required to find the distance, not the velocity, so need more integration in order.

$$s = \int v\,dt = \int (-8t + 24)dt$$

$$s = -4t^2 + 24t + c_2$$

To determine c_2, the stopping distance s is measured from where, and when, the brakes are applied so that at $t = 0,\ s = 0$.

$$s = -4t^2 + 24t + c_2 \Rightarrow 0 = -4(0)^2 + 24(0) + c_2 \Rightarrow c_2 = 0$$

$$s = -4t^2 + 24t$$

The stopping distance s could be evaluated if we knew the braking time. The time can be determined from the speed statement.

The bike stops when $v = 0$, $\Rightarrow v = -8t + 24 \Rightarrow 0 = -8t + 24 \Rightarrow t = 3$.

When $t = 3$, we get

$$s = -4t^2 + 24t \Rightarrow s = -4(3)^2 + 24(3)$$

$$s = 36 \text{ metres} < 40 \text{ metres}$$

The bike stops at a distance 4 metres to the barrier.

EXERCISE 11.4

(1) If $f'(x) = 4x - 5$ and $f(2) = 1$, find $f(x)$.

(2) If $f'(x) = 9x^2 - 6x$ and $f(0) = -3$, find $f(x)$.

(3) If $f''(x) = 12x - 6$ and $f(1) = 30$, $f'(1) = 5$ find $f(x)$.

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